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Student Number

2016

Mathematics

HSC Course Assessment Task 3

1st June 2016

- General Instructions**
- Reading time – 5 minutes
 - Working time **60 minutes**
 - Write using blue or black pen
Black pen is preferred
 - Board approved calculators may be used
 - In Questions 4 - 16 show relevant mathematical reasoning and/or calculations
 - Use a separate booklet for each question
 - A reference sheet is provided at the back of this paper
 - Diagrams are NOT to scale (unless indicated)

Total Marks – 42

- Section I** - Page 2
- 3 marks**
- Attempt Questions 1 - 9
 - Allow about 5 minutes for this section

- Section II** - Pages 3 - 5
- 41 marks**
- Attempt Questions 4 - 6
 - Allow about 55 minutes for this section

Multiple Choice	/ 3
Q 4 (Logarithms and Exponentials)	/ 21
Q 5 (Probability)	/ 7
Q6 (Applications of Calculus to the Physical World)	/ 11
Total	/ 42

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 20% of the HSC Course Assessment

Section I

3 marks

Attempt Questions 1 - 3

Allow about 5 minutes for this section

Use the multiple-choice answer sheet for questions 1 -3 (Detach from paper)

1. Given that $\log_a x = 0.3$ and $\log_a y = 0.2$, then the value of $\log_a x^2 y^3$ is
- (A) $a^{0.36}$
(B) 1
(C) 1.2
(D) $a^{0.0036}$
2. If $y = e^x - e^{-x}$, then the gradient of the tangent at $x = 0$ is:
- (A) 0
(B) 2
(C) -2
(D) e^2
3. At Killara High School the probability that a student takes a Mathematics course is 0.9 and the probability that a student takes a Science course is 0.8. The probability that a student takes a Mathematics or a Science course is 0.95.

What is the probability that a student does both Mathematics and Science?

- (A) 0.1
(B) 0.2
(C) 0.75
(D) 1

Section II

39 marks

Attempt Questions 4 - 6

Allow about 55 minutes for this section

Answer each question in a separate booklet. Extra writing booklets are available.

In questions 4 – 6 your responses should include relevant mathematical reasoning and / or calculations.

Question 4 (21 marks) Use a SEPARATE writing booklet

(a) Differentiate with respect to x and fully simplify

i) $y = e^x \log_e x$ 2

ii) $y = \cos(\log_e x)$ 2

(b) Evaluate using exact values $\int_0^{e^2} \frac{4}{x} dx$ 3

(c) Solve for x 3
$$e^{2x} - 2e^x = 0$$

(d) Let $y = \log_e(x - 1)$

i) Find the x – intercept. 2

ii) Sketch the graph of $y = \log_e(x - 1)$. 2

iii) Find the exact value of the area bounded by $y = \log_e(x - 1)$, x – axis and the line $x = 5$. 4

iv) Find the exact volume of the solid generated when the area between $y = \log_e(x - 1)$, x – axis and y – axis is rotated about the y – axis. 3

Question 5 (7 marks) Use a SEPARATE writing booklet

- a)** Two standard six sided dice are tossed. Given that a 4 is showing on the upmost face of one of the dice , what is the probability that the sum of the upmost faces on both dice exceeds 8? **2**

- b)** In a game of chess between two players, X and Y, of about equal ability, the player with the white pieces, having the first move, has a probability of 0.5 of winning, and the probability that the player with the black pieces wins is 0.3.

The two players, X and Y, play each other twice in a chess competition, each player having the white pieces once. In the competition, a player who wins a game scores 1 point, a draw scores as $\frac{1}{2}$ a point to each player and a player who loses scores 0 points.

- i)** What is the probability that the game ends in a draw (i.e. no player wins) **1**
- ii)** What is the probability that, as a result of these two games X scores 2 points? **2**
- iii)** What is the probability that, as a result of these two games X scores $1\frac{1}{2}$ points? **2**

Question 6 (11 marks) Use a SEPARATE writing booklet

- a)** The rate of flow (R litres per minute) of water through a filter is given by

$$R = 6t - \frac{t^2}{3}$$

- i)** At what times is no water flowing through the filter **2**
- ii)** Find the volume of the water that flows through the filter in the first 4 minutes **3**

- b)** In a chemical reaction the amount (M kilograms) of undissolved solid after t hours is given by

$$M = Ae^{-kt}.$$

After 24 hours 15% of the initial mass of solid remains undissolved.

- i)** The value of k **2**
- ii)** What percentage of solid has dissolved after 5 hours (give your answer correct to one decimal place) **2**
- iii)** How long it takes for 50% of the solid to be dissolved. (give your answer correct to the nearest minute) **2**

End of paper ☺

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Student Number



HSC Mathematics

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 3. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 A ☒ B ☒ C ☐ D ☐

Start Here



1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐

HSC Mathematics Task 3 2016 Marking Scheme

Question 4

(a) Differentiate with respect to x and fully simplify

i) $y = e^x \log_e x$

2

1 mark – correctly identifies need to use product rule 1 mark – finds the correct derivative from product rule	
$y = e^x \log_e x$ $u = e^x \quad v = \log_e x$ $u' = e^x \quad v' = \frac{1}{x}$ $\therefore \frac{dy}{dx} = uv' + vu'$ $\therefore \frac{dy}{dx} = \frac{e^x}{x} + e^x \log_e x$	<i>Generally very well done</i>

ii) $y = \cos(\log_e x)$

2

1 mark – gives $\sin(\log_e x)$ as the solution or a minor error 2 mark – finds the correct derivative from chain rule	
$y = \cos(\log_e x)$ $\frac{dy}{dx} = -\sin(\log_e x) \times \frac{1}{x} = -\frac{\sin(\log_e x)}{x}$	<i>Generally very well done</i>

(b) Evaluate using exact values $\int_1^{e^2} \frac{4}{x} dx$

3

1 mark – finds correct primitive 1 mark – substitutes limits correctly 1 mark – finds correct exact value	
$\int_1^{e^2} \frac{4}{x} dx = 4[\ln x]_1^{e^2}$ $= 4[\ln e^2 - \ln 1]$ $= 4[2 - 0]$ $= 8$	<i>Some students got mixed up with the constant – but generally well done</i> <i>May did not simplify to the whole number</i>

(c) Solve for x

3

$$e^{2x} - 2e^x = 0$$

1 mark – factorises as a quadratic

1 mark – discounts $e^x = 0$ with a reason

1 mark – correct solution $x = \log_e 2$

$$e^{2x} - 2e^x = 0$$

$$\therefore e^x(e^x - 2) = 0$$

$$\therefore e^x = 0 \text{ no solution since } e^x > 0 \text{ OR } \therefore e^x = 2$$

$$x = \log_e 2$$

Many students did not create a quadratic

Many students did not say why the solution $e^x = 0$ and lost a mark

(d) Let $y = \log_e(x - 1)$

i) Find the x – intercept.

2

1 mark – correct x intercept

1 mark – sufficient working

$$y = \log_e(x - 1)$$

$$\therefore 0 = \log_e(x - 1)$$

$$\therefore x = 2$$

x intercept is $(2, 0)$

Genearily very well done

ii) Sketch the graph of $y = \log_e(x - 1)$

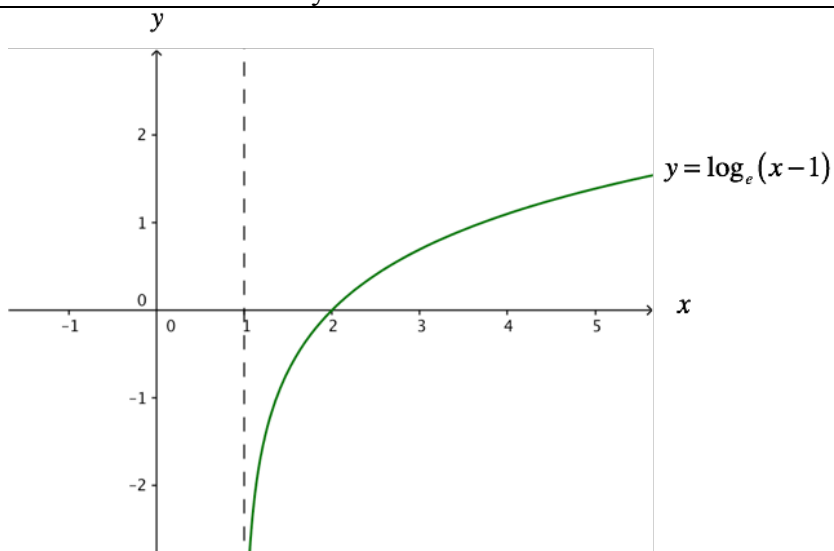
2

1 mark – x intercept and asymptote clearly and correctly indicated

1 mark – correct shape of graph

lose 1 mark if not evenly marked scale

Many did not label the asymptote or have the correct one
Many had no even scale or their graph did not work on the scale they had given



- iii) Find the exact value of the area bounded by $y = \log_e(x-1)$, x -axis and the line $x = 5$.

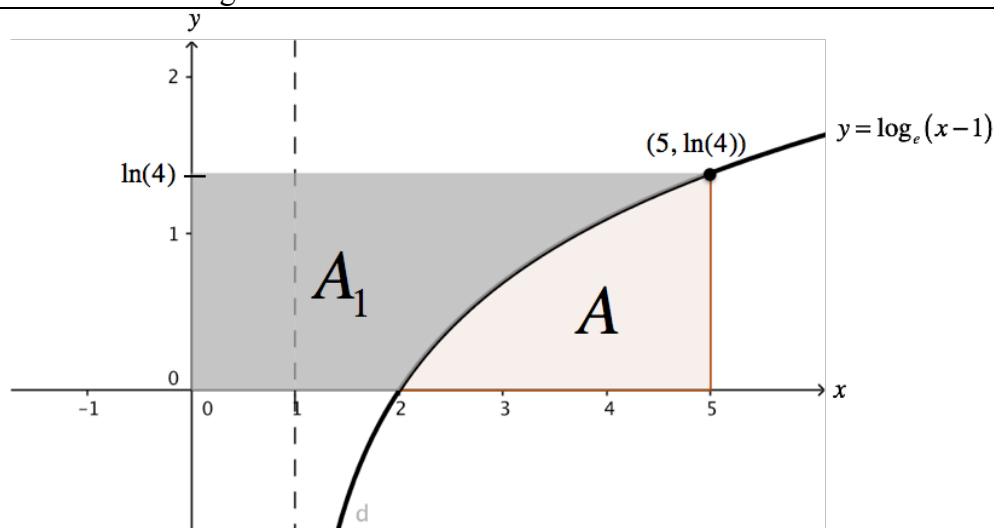
4

1 mark – breaking into rectangle with two separate areas

1 mark - making x the subject if $y = \log_e(x-1)$
 $\therefore x = e^y + 1$

1 mark – finding correct primitive and expression for the area

1 mark - finding exact value of area



*Not very well done
 Many tried to integrate log
 Some found A_1 but did not
 subtract from rectangle*

if $y = \log_e(x-1)$

$\therefore x = e^y + 1$

$$\int_2^5 \log_e(x-1) dx = A_{\text{rectangle}} - A_1$$

$$= (\ln(4) \times 5) - \int_0^{\ln 4} e^y + 1 dy$$

$$= 5 \ln(4) - [e^y + y]_0^{\ln 4}$$

$$= 5 \ln 4 - [(4 + \ln 4) - (1 - 0)]$$

$$= (4 \ln 4 - 3) \text{ units}^2$$

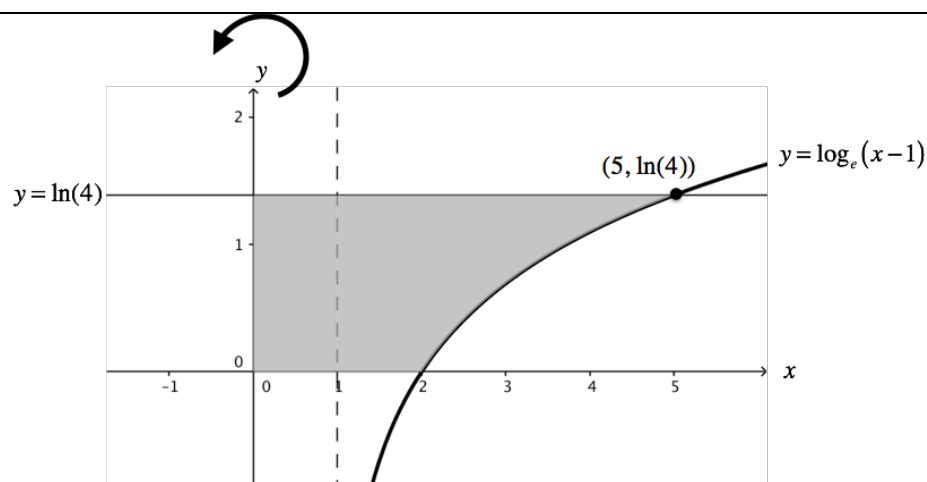
- iv) Find the exact volume of the solid generated when the area between $y = \log_e(x-1)$, the line $y = \log_e 4$, the x -axis and the y -axis is rotated about the y -axis.

3

1 mark: $V = \pi \int_0^{\ln 4} (e^y + 1)^2 dy$

1 mark: correct primitive for their integral (unless trivial)

1 mark: correct value for their integral



Many did not square or change the subject

Many did not square correctly

Generally poorly done

$$\begin{aligned}
 V &= \pi \int_0^{\ln 4} x^2 dy \\
 &= \pi \int_0^{\ln 4} (e^y + 1)^2 dy \\
 &= \pi \int_0^{\ln 4} (e^{2y} + 2e^y + 1) dy \\
 &= \pi \left[\frac{1}{2} e^{2y} + 2e^y + y \right]_0^{\ln 4} \\
 &= \pi \left[\left(\frac{e^{2\ln 4}}{2} + 2e^{\ln 4} + \ln 4 \right) - \left(\frac{1}{2} + 2 + 0 \right) \right] \\
 &= \pi \left[\left(\frac{16}{2} + 2 \times 4 + \ln 4 \right) - \left(\frac{5}{2} \right) \right] \\
 &= \pi \left(\frac{27}{2} + \ln 4 \right)
 \end{aligned}$$

Question 5

- a) Two standard six sided dice are tossed. Given that a 4 is showing on the upmost face of one of the dice, what is the probability that the sum of the upmost faces on both dice exceeds 8? 2

1 mark – lists the sample space in any format 1 mark – finds the correct probability																																																									
The sample space for a 4 showing on one upmost face Dice 1 <table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Dice 2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td></td><td></td><td></td><td>5</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td><td>6</td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td><td>7</td><td></td><td></td></tr><tr><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>5</td><td></td><td></td><td></td><td>9</td><td></td><td></td></tr><tr><td>6</td><td></td><td></td><td></td><td>10</td><td></td><td></td></tr></table> $\therefore P(\text{exceeds } 8) = \frac{4}{11}$		1	2	3	4	5	6	Dice 2							1				5			2				6			3				7			4	5	6	7	8	9	10	5				9			6				10			
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- b) In a basketball shootout, a player gets one attempt at scoring a 3 point shot and three attempts at scoring from a one point shot. From past statistics a players has a probability of 0.2 of scoring a three point shot and a probability of 0.75 of scoring a one-point shot.

- i) What is the probability that a player scores 0 points 1

1 mark: correct probability with working	
$P(0 \text{ points}) = 0.8 \times (0.25)^3 = 0.0125$	

- ii) What is the probability that a player scores all three one point shots but misses the three point shot? 1

1 mark: correct probability with working	
$P(\text{score three 1 points but miss 3 point}) = 0.8 \times (0.75)^3 = 0.3375$	

- iii) What is the probability that, a player scores at least 2 points? 3

1 mark: recognises complimentary event or lists possible outcomes 1 mark: correctly calculates the probability for 1 point 1 mark: obtains correct probability	
$ \begin{aligned} P(\text{at least 2 points}) &= 1 - (P(0 \text{ points}) + P(1 \text{ point})) \\ &= 1 - (0.0125 + (0.8 \times 0.75 \times (0.25)^2 \times 3)) \\ &= 0.959 \end{aligned} $	

Question 6

- a) The rate of flow (R litres per minute) of water through a filter is given by

$$R = 6t - \frac{t^2}{3}$$

- i) At what times is no water flowing through the filter

2

1 mark: equates derivative to 0 1 mark: correctly finds both times	
we need $R = \frac{dV}{dt} = 6t - \frac{t^2}{3} = 0$ $\therefore 18t - t^2 = 0$ $\therefore t(18 - t) = 0$ $\therefore t = 0$ minutes or $t = 18$ minutes	

- ii) Find the volume of the water that flows through the filter in the first 4 minutes

3

1 mark: establishes correct integral 1 mark: finds correct primitive function 1 mark: finds correct value and gives units	
Since $\frac{dV}{dt} > 0$ for $0 \leq t \leq 4$ $V(\text{water first 4 min}) = \int_0^4 \left(6t - \frac{t^3}{3} \right) dt$ $= \left[3t^2 - \frac{t^4}{12} \right]_0^4$ $= \left[3(4)^2 - \frac{4^4}{12} \right] - [0]$ $= \frac{368}{3} \text{ litres}$	

- b)** In a chemical reaction the amount (M kilograms) of undissolved solid after t hours is given by

$$M = Ae^{-kt}.$$

After 24 hours 15% of the initial mass of solid remains undissolved.

- i)** The value of k

2

1 mark: substituting into expression to obtain $0.15A = Ae^{-24k}$ 1 mark: finds correct value of k (any number of decimal places) with working	
When $t = 24$, $M = 0.15A$ $\therefore 0.15A = Ae^{-24k}$ $\therefore 0.15 = e^{-24k}$ $\therefore \log_e(0.15) = -24k$ $\therefore k = -\frac{\log_e(0.15)}{24}$ $\therefore k = 0.07904666604$	

- ii)** What percentage of solid has dissolved after 5 hours (give your answer correct to one decimal place)

2

1 mark: finding $\therefore M = 0.6735228677A$ with working 1 mark: correctly concluding the amount of the original solid that has dissolved	
When $t = 5$, $k = -\frac{\log_e(0.15)}{24}$, $M = ?$ $\therefore M = Ae^{-5 \times \left(\frac{\log_e(0.15)}{24}\right)}$ $\therefore M = 0.6735228677A$ (this is the amount remaining) $\therefore 32.65\%$ of the original solid has dissolved	

- iii)** How long it takes for 50% of the solid to be dissolved. (give your answer correct to the nearest minute)

2

1 mark: finding $\therefore \log_e(0.5) = -\left(\frac{\log_e(0.15)}{24}\right)t$ 1 mark: correctly identifying the correct time (to the nearest minute)	
When $t = ?$, $k = -\frac{\log_e(0.15)}{24}$, $M = 0.5A$ $\therefore 0.5A = Ae^{-\left(\frac{\log_e(0.15)}{24}\right)t}$ $\therefore \log_e(0.5) = -\left(\frac{\log_e(0.15)}{24}\right)t$ $\therefore t = 8$ hours 46 minutes	